

Doomersion

The hook that forgot to land.

Company	Doomersion · Doomlingo Inc. · San Francisco
Backing	Y Combinator Winter 2026
Team	4-person team · Founder: Mostafa Afr, age 23
Launch Date	March 9, 2026 · Written: May 2026 (10 weeks post-launch)
Downloads	10,000+ (Google Play) · 47 App Store ratings · 5 written reviews
Frameworks	JTBD · KANO · ICE Scoring · 5 Whys RCA · Effort-Impact Matrix
Research	Primary: 20 min personal usage · Secondary: 5 App Store reviews (Feb–Apr 2026)
Author	Aman Raj Singh · Day 25 of 31 · PM Transition · Building in Public

This document presents the complete analytical framework for Case Study 04, structured across nine layers: company context and competitive positioning, user research and evidence clustering, problem statement and 5 Whys root cause analysis, KANO prioritisation, user stories, solutions with constraints and trade-offs, ICE scoring, effort-impact matrix, A/B test design, success metrics, risks and mitigations, product roadmap, and vision narrative.

The Bet Doomersion Made

Doomersion's thesis is simple and differentiated: people already spend hours doomscrolling. Instead of fighting that habit, route it into language learning. Short videos in the user's target language, matched to their level, getting harder as they scroll. The acquisition hook is genuinely novel — not another streak-based app, not another flashcard system. A feed.

Company Snapshot

Backing	Y Combinator Winter 2026
Team Size	4 people. Founder Mostafa Afr, age 23, Penn graduate, former professional gamer.
Launch Date	March 9, 2026. This teardown written 10 weeks post-launch.
Downloads	10,000+ on Google Play. No written reviews yet. App Store: 47 ratings, 5 written reviews.
Product Stage	0→1. Must Haves not yet solid. Activation is the core unsolved problem.
Business Model	Freemium. Free tier: basic feed and word lookup. Paid: AI assistant, analytics, offline.

Jobs To Be Done (JTBD)

Users are not hiring Doomersion to 'learn a language.' They are hiring it to make their screen time feel productive without changing the habit. That is a fundamentally different job. One leads to Duolingo. The other leads to something nobody has built yet at scale.

Competitive Positioning

Duolingo	Structured daily learning	Too gamified; feels like homework to Gen Z users
YouTube	Long-form language immersion	Too passive; no learning scaffold or feedback loop
TikTok	Scroll habit and attention	No learning layer whatsoever
Doomersion	The intersection of all three	Contextual learning inside the scroll — unbuilt at scale

What Users Actually Experienced

Research Transparency Note

Google Play Store: zero written reviews despite 10,000+ downloads. App Store: 47 ratings, 5 written reviews across February–April 2026. Primary research: 20 minutes of personal app usage documented in real time. Reddit, Product Hunt, G2, YC launch page: no relevant user comments found. Sample is small and this limitation is acknowledged throughout. Given the limited secondary sources available, primary usage findings carry equal weight to review data.

Primary User Segment Identified

Gen Z college student, aged 18–25, heavy Instagram and TikTok user, wants their screen time to mean something without it feeling like structured study again. Evidence signals: app name references doomscrolling, founder gaming background, App Store review language, and product positioning copy.

Evidence Clusters

Cluster	KANO	Source	Pattern
Performance & Technical Bugs	Must Have	Personal + 3 reviews	Lag kills the TikTok-like experience. Arabic transcripts render as one long word. Rewind crashes video. Captions delayed.
Content Language Mismatch	Must Have	Personal + 2 reviews	Chinese learner shown Spanish-dominant video. Dual foreign language confusion. Algorithm tags word not video language.
Broken Onboarding	Must Have	Personal only	No guidance on core loop. 4 videos before signup wall. Clickable word has no visible contextual bridge to video.
Content Variety Too Narrow	Performance	3 reviews	Only skits, comedy, dance. No search. No algorithm influence. Boring for less common languages.

What Works vs What Does Not

What Works	What Does Not Work
Acquisition hook — "learn by doomscrolling" genuinely differentiated	Activation — users do not understand the core loop without guidance
Word tap feature — when discovered, clicking a word and seeing breakdown works well	Content matching — wrong language videos shown to declared target learners

Quiz feature — appreciated by the users who reach it	AI feature — 3 consecutive error responses in personal testing
Core concept — power users report 3-hour sessions vs Duolingo 5-minute average	Performance — lag on a short-form video app is fatal; TikTok is the comparison point

One Sentence Explains the Churn

Problem Statement

A Gen Z user, tired of mindless doomscrolling on Instagram or TikTok, hired Doomersion to make their screen time feel productive — learning a language without changing the habit. Instead they encountered a broken first experience: no guidance on how the app works, videos with content in unrelated languages, lag that kills the feed, and a UI that demands too much cognitive effort to follow. This costs Doomersion the one thing a 4-person YC startup cannot afford to lose — Day 1 to Day 7 retention — leaving a brilliant acquisition hook with no activation to back it up.

5 Whys Root Cause Analysis

Why 1	Users churn before understanding the product's value.
Why 2	Because the first experience is confusing — lag, no onboarding guidance, and videos in unrelated languages make the core loop invisible.
Why 3	Because the feed serves content that does not match the user's declared target language, breaking context before learning can begin.
Why 4	Because the content curation algorithm pulls videos based on the target language word appearing somewhere in the video — not based on whether the video itself is in or about that language.
Why 5	Because the app has no backend audio processing to understand what language is actually being spoken. It relies on surface-level tagging. Proactive audio segmentation and contextual subtitle generation were not prioritised in the 0→1 build.

Root Cause

"Doomersion's algorithm places words in context but not videos in context. It finds the right word but puts it inside the wrong world — leaving the user linguistically lost before the learning even starts."

What to Fix First and Why

Doomersion is a 10-week old product with a 4-person team. KANO principle for early products: fix all Must Haves before any Performance feature. No Delighters until the foundation is solid.

KANO Framework Applied

Must Have — Fix First	Performance — Fix After	Delighters — Later
Feed Language Intelligence Wrong language kills trust before learning starts.	Lag & Video Quality Fatal but not the first churn reason.	Social features — cohort learning, leaderboards
Progressive Onboarding Without guidance, activation fails entirely.	Feed Personalisation Builds habit but requires Must Haves first.	Corporate-standard learning certificates
Context-Aware AI Broken AI destroys the second core promise.		Creator tools — native speaker content

User Stories

MH1 — Feed Language	As a college student who spends hours on Instagram Reels, I want to absorb a new language through videos of real people from that culture, so that my screen time feels productive without feeling like structured study.
MH2 — Onboarding	As a college student used to the effortless smoothness of Instagram Reels, I want to start learning a new language without any setup confusion, so that my free time feels enhanced — not spent figuring out a new app.
MH3 — Context AI	As a college student used to asking doubts fast using LLMs like ChatGPT, I want AI language support seamlessly inside the app, so that I do not have to switch to ChatGPT and break my attention.
P1 — Performance	As a college student used to instant-loading feeds, I want a smooth experience without waiting or frustration, so that my learning time feels effortless and I do not switch back to Instagram or TikTok.
P2 — Feed Algorithm	As a college student with a highly personalised Instagram and YouTube feed, I want my learning feed to understand my viewing patterns and show varied content, so that the app feels intelligent and relevant — not generic.

Five Interventions, One Direction

Solution 1 · Must Have · Core Algorithm

Feed Language Intelligence

Backend AI processes every video before it reaches the feed — detecting spoken language through audio analysis and visual cues. Only videos matching the user's declared target language enter the feed. For matched videos, real-time sentence-level subtitles appear below in the target language. Tapping any sentence pauses the video and shows word-by-word translation with context.

Constraint	Backend processing increases feed load latency. Requires powerful compute infrastructure — paid cloud processing and model training costs are significant for a 4-person team.
Trade-off	Choosing server-side language filtering over on-device processing — more accurate but adds latency. Speed is sacrificed for reliability at this stage.
Business Value	Improves Day 1 Activation Rate — % of new users who complete their first full subtitle interaction.
A/B Test	50% new users receive current feed; 50% receive language-filtered feed. Measure: Day 1 activation rate. Success threshold: 20%+ improvement in filtered group after 500 users per group.

Solution 2 · Must Have · Quick Win

Progressive Onboarding

After initial questions, user lands on feed with a guided first experience. Step 1: nudge to tap the subtitle word. Step 2: nudge to swipe up/down. After 2–3 videos: Gen Z-style prompt to try AI feature. After AI used: 2 more free scrolls then signup wall. Only these steps. Main feed visible in background throughout — user experiences the product while learning to navigate it.

Constraint	Risk is purely in nudge design. Intrusive nudges backfire. Requires careful UX testing: visible enough to guide, light enough not to frustrate.
Trade-off	Choosing linear guided onboarding over interactive tutorial — fewer steps means less drop-off, even though some features remain undiscovered initially.
Business Value	Improves Day 1 to Day 3 Retention Rate — % of users who return within 72 hours of first install.
A/B Test	50% receive current experience; 50% receive guided onboarding. Measure: Day 3 retention. Success: 15%+ higher return rate. Fastest test to conclude — 72 hours per user.

Context-Aware AI

AI tracks which part of the app it was opened from and understands user intent from context. Instead of generic error messages, it delivers short constructive redirects — telling the user specifically what to try instead of presenting a dead end. When the AI cannot process a request, it responds with a specific useful suggestion.

Constraint	Every AI call must carry session context — current screen, user intent, recent actions — significantly increasing token usage per request.
Trade-off	Choosing context-aware short responses over a full conversational tutor — reliability matters more than depth at this stage. A simple AI that always responds beats a sophisticated AI that errors out.
Business Value	Increases AI Feature Engagement Rate — % of active users who use the AI feature at least once per session.
A/B Test	50% receive current AI; 50% receive context-aware AI. Measure: AI engagement rate per session. Secondary metric: session length. Success: 25%+ higher AI usage in new group.

Performance and Lag Optimisation

Declutter feed UI to reduce rendering load. Implement adaptive bitrate streaming — video quality adjusts dynamically to available bandwidth so users on 3G experience no lag. Apply AI-assisted video smoothing capped at a defined pixel threshold, preserving original content integrity while improving perceived quality on lower-end devices.

Constraint	AI video smoothing at scale multiplies compute costs across concurrent users. Adaptive bitrate streaming is the more cost-effective approach for a 4-person team.
Trade-off	Choosing adaptive bitrate streaming over AI smoothing — proven technology, lower cost, does not multiply per-user compute costs at scale.
Business Value	Reduces Day 1 Churn Rate — % of users who do not return within 24 hours of first session.
A/B Test	50% current video delivery; 50% optimised adaptive bitrate. Measure: Day 1 churn. Segment specifically by lower-end Android devices and 3G connections for the sharpest signal.

Personalised Feed Algorithm

Dynamic recommendation engine tracks watch time per video in real time. If a user spends more time on a content type — identified through subtitle keywords and video hashtags — push similar content. If engagement drops across 3–4 consecutive videos, reset to general feed and recalculate. Force 7–8 general videos periodically even when engagement is strong, to prevent filter bubble fatigue. Learns within session first, then across sessions.

Constraint	Algorithm development time is significant. Edge cases are complex: passive scrollers, mixed intent users, cold start problem for new users with no watch history.
Trade-off	Choosing watch-time as the primary signal over a multi-signal algorithm — simplest reliable signal available. Hashtags and subtitles add complexity a small team cannot maintain cleanly in v1.
Business Value	Improves Day 7 and Day 30 Retention Rate — % of users still active one week and one month after install.
A/B Test	50% static feed; 50% dynamic watch-time algorithm. Measure: Day 7 and Day 30 retention. Longest test window — 30 days minimum. Run in parallel with faster tests.

Scoring the Build Order

ICE scores each solution on three dimensions: Impact (how much does this move Day 1–7 retention, 1–10), Confidence (how sure are you based on evidence, 1–10), and Ease (how easy for a 4-person team, where 10 = very easy, 1–10). Final score = Impact × Confidence × Ease.

Solution	KANO	Impact	Confidence	Ease	ICE Score
P1 — Performance & Lag	Performance	8	8	7	448
MH1 — Feed Language Intelligence	Must Have	8	7	5	280
MH2 — Progressive Onboarding	Must Have	7	3	10	210
MH3 — Context-Aware AI	Must Have	5	4	7	140
P2 — Personalised Feed Algorithm	Performance	6	4	3	72

Key insight: P1 scores highest on ICE but is a Performance feature. KANO overrides — all Must Haves ship first regardless of ICE score. MH2 is the single highest priority action: it is both a Must Have AND the easiest to build (Ease: 10). A 4-person team should ship this in days.

Effort vs Impact Matrix

Do First — High Impact, Low Effort	Plan — High Impact, High Effort
MH2 — Progressive Onboarding P1 — Performance & Lag	MH1 — Feed Language Intelligence P2 — Personalised Feed Algorithm
Fill In — Low Impact, Low Effort	Avoid — Low Impact, High Effort
MH3 — Context-Aware AI	Nothing. A focused solution set.

How You Know It Is Working

Each solution has one primary metric. Baseline and target numbers are not stated as internal data is unavailable; timeframes are defined based on the nature of each metric.

MH1	Day 1 Activation Rate	% of new users who complete first full subtitle interaction — tap a word and see translation.	Within first 24 hours of install
MH2	Day 1 to Day 3 Retention Rate	% of users who return and open the app within 72 hours of first install.	Measured over first 72 hours post-install
MH3	AI Feature Engagement Rate	% of active users who use AI feature at least once per session. Secondary: session length.	Measured weekly after feature ships
P1	Day 1 Churn Rate	% of users who do not return within 24 hours of first session.	Within first 24 hours of install
P2	Day 7 and Day 30 Retention	% of users still active one week and one month after install. Day 7 is leading; Day 30 confirms habit.	Day 7 at week end; Day 30 at month end

Leading indicators (Day 1, Day 3) provide fast iteration signals within days of shipping. Lagging indicators (Day 7, Day 30) confirm long-term impact but require patience. Run A/B tests on leading indicators first; use lagging indicators to validate sustained results.

What Could Go Wrong

MH 1	Feed Load Latency	Backend processing of every video increases time before content reaches the feed. On a short-form video app, any perceptible load time breaks the scroll experience.	Pre-filter content on first user profiling to build a cached content library. Give the real-time system a head start so the user never feels the processing delay.
MH 2	Onboarding Drop-off	Too many steps before the main screen causes the user to feel onboarding is the product. They came to scroll, not to be taught how to scroll.	Show the main feed in the background between nudge steps. User experiences the product while learning to navigate it — service delivery and education happen simultaneously.
MH 3	Token Cost Multiplication	Context-aware AI requires passing full session state with every API call, significantly increasing token usage and backend cost per active user.	Limit AI access to specific defined touchpoints in the app. Shorter context per call keeps costs controlled without sacrificing contextual intelligence.
P1	Content Integrity Compromise	AI video smoothing risks altering the original video content. Over-smoothed footage loses authenticity, which matters particularly for cultural and language immersion.	Backend caps smoothing at a defined pixel threshold. Preserves original content integrity while improving perceived quality. Never apply above the cap regardless of device capability.
P2	Filter Bubble Fatigue	Watch-time algorithm creates a content echo chamber. Users get locked into one content type, eventually exhausting that category and churning faster than a varied feed would cause.	Force 7–8 general videos into the feed periodically even when the engagement trend is positive. This resets and realigns user interest, preventing the filter bubble from forming.

Where Doomersion Could Go

6-Month Product Roadmap

Now 0–3 months	Fix the Foundation	Ship all three Must Haves. Progressive onboarding first — days to build, immediate retention impact. Language feed filtering and context-aware AI in parallel. Performance optimisation running alongside.
Next 3–6 months	Build the Habit Loop	Ship personalised feed algorithm. Introduce streak mechanics and vocabulary tracking across sessions. Build the repeat-usage trigger that converts curious downloaders into daily learners.
Later 6–12 months	Build the Ecosystem	Validated learning credentials — corporate-standard certificates proving what the user learned. Doomersion becomes not just a learning app but a credential platform. Screen time converted to career proof.

Vision Narrative — Alex

Alex is 21. Final year engineering in Pune. His Instagram feed knows him better than his friends do — travel reels, tech memes, Japanese street food videos he cannot stop watching.

One night he sees a post about Doomersion. 'Learn Japanese by doomscrolling.' He downloads it at 11pm thinking he will try it for five minutes.

The app asks him three questions. Target language. Level. Done. The feed opens and a small nudge appears at the bottom — not a tutorial, just one instruction: 'Tap any word below the video.' He taps. The video pauses. The word breaks apart: kanji, reading, meaning. He did not feel like he studied. He felt like he caught something.

Two videos later another nudge. 'Swipe left to ask about what you just watched.' He swipes. The AI asks him a specific question about the scene. He types back. It responds in two lines. Fast. He did not open ChatGPT. He did not break his scroll.

Three weeks later Alex is 40 videos deep into his Japanese streak. His feed has figured him out — street food videos, anime reactions, Tokyo vlogs. All Japanese. All at his level. Getting harder without him noticing.

Then something new appears. A badge: 'Conversational Beginner — Doomersion Verified.' A short test. Ten videos. Ten word questions. He passes. He puts it on his LinkedIn. His friend messages him: 'What is this?' He sends the download link.

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